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User Interface Design

Field	Value
Document ID	NOVA-UI-001
Version	0.1
Status	Draft
Owner	Laxmi Poudel
Date	2026-09-08
Conforms to	Project convention. Screen inventory and low-fidelity wireframes

1. Purpose and scope

NOVA-SRS-001 section 3.4.1 names four views and says nothing further about them. This document closes gap G3 in NOVA-SDP-001 section 5.6 by fixing what each view contains, which requirement each element serves, and what each view shows when things go wrong.

Two of these views carry the project's explanatory value. The trace viewer is the evidence that the system can be inspected rather than trusted, and the applied-lessons panel is the evidence that memory is doing anything at all. A demonstration in which neither is legible has nothing to show.

In scope. Screen inventory, layout structure, element-to-requirement traceability, state coverage including every exception condition in NOVA-SRS-001 section 3.5.2, and navigation.

Out of scope. Visual design, typography, colour, component library selection, and copy. Those are implementation decisions and are not load-bearing for any requirement.

Wireframes are deliberately low-fidelity ASCII. They are diffable, they survive document generation, and they cannot be mistaken for a finished design.

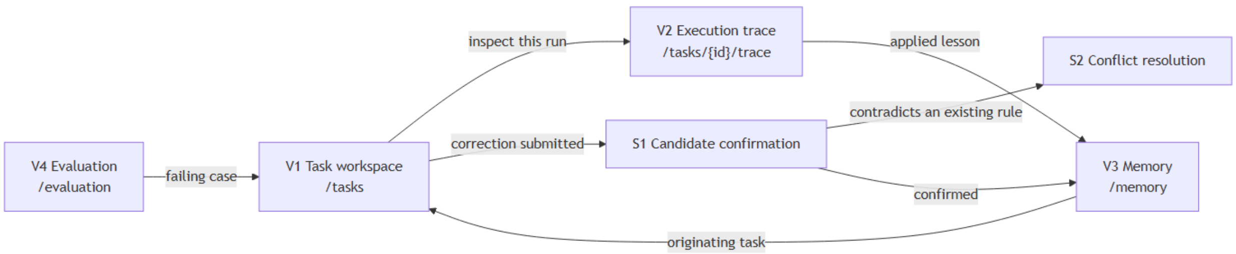
2. Screen inventory

ID	View	Route	Purpose	Primary requirements
V1	Task workspace	`/tasks`, `/tasks/{id}`	Submit a requirement, watch it execute, read the plan, record feedback	FR-1, FR-2, FR-5, FR-4.4, NFR-7, NFR-8
V2	Execution trace	`/tasks/{id}/trace`	Show every step, score, model call, and check verdict behind one plan	FR-8 to FR-8.4
V3	Memory management	`/memory`, `/memory/{id}`	Browse, filter, edit, pin, delete, and export stored rules	FR-7 to FR-7.7
V4	Evaluation results	`/evaluation`	Report metrics per run, and show the gate blocking a configuration	FR-9, FR-9.1, FR-9.2, FR-10.1

Two surfaces are not separate views but are called out because they carry requirements of their own:

ID	Surface	Host view	Purpose	Primary requirements
S1	Candidate confirmation	V1	Approve, edit, or discard a rule derived from a correction	FR-6.1 to FR-6.4, FR-6.7
S2	Conflict resolution	S1	Present a candidate and the in-scope memory it contradicts	FR-6.5, OP-4

2.1 Navigation



Every navigation edge is bidirectional in practice through the browser's own history. No view is reachable only from one other view.

3. Persistent frame

Present on every view.

```
+------------------------------------------------------------------------------+
| NOVA        Tasks    Memory    Evaluation           provider: local * gemma3:4b  |
+------------------------------------------------------------------------------+
|                                                                              |
|  [ view body ]                                                               |
|                                                                              |
+------------------------------------------------------------------------------+
```

The provider indicator satisfies FR-11.2 and is a permanent element rather than a settings-page detail, because "did this leave my machine" is a question a user should never have to navigate to answer. It states the provider and the model. Where the provider is not local it is stated in words, never by absence or by colour alone.

4. V1. Task workspace

4.1 Submission

```
+------------------------------------------------------------------------------+
| New task                                                                     |
|                                                                              |
| Requirement type   [ API / CRUD endpoint            v ]                      |
|                                                                              |
| Requirement                                                                  |
| +--------------------------------------------------------------------------+ |
| | As an admin, I can delete a user account via DELETE /users/{id}.         | |
| |                                                                          | |
| | AC1: ...                                                                 | |
| +--------------------------------------------------------------------------+ |
|                                                 1,284 / 8,000 characters     |
|                                                                              |
| Supplementary context (optional)                                             |
| +--------------------------------------------------------------------------+ |
| |                                                                          | |
| +--------------------------------------------------------------------------+ |
```

```
|                                                                          |
|                                                       [ Generate plan ]  |
+--------------------------------------------------------------------------+
```

The character counter is present before submission rather than after, because FR-1.1 rejects an over-length requirement and a limit discovered by rejection is a limit discovered too late.

4.2 Execution

Submission returns a task identifier without waiting (section 3.5.1 step 2), so this state is reached immediately and is polled.

```
+--------------------------------------------------------------------------+
| Task 0f3a  Â·  Delete user  Â·  RUNNING                        [ Cancel ]    |
|                                                                          |
|  Queued 14:02:11  ->  Running 14:02:12  ->  step 3 of 5: generating      |
|                                                                          |
|  retrieval    done    3 memories applied, 1 excluded below threshold     |
|  selection    done    playbook "endpoint-standard"                       |
|  generation   ...     attempt 1                                          |
|  checks       -                                                          |
|  review       -                                                          |
|                                                                          |
|                                               [ Inspect trace so far ]   |
+--------------------------------------------------------------------------+
```

Status is a word, not a colour or a spinner alone (NFR-7). Cancel is present for the whole of the running state and preserves the partial trace (NFR-8, OP-9). The trace is reachable while the task is still running, because a task that hangs is exactly when its trace is most wanted (FR-8.3).

4.3 Results and feedback

```
+--------------------------------------------------------------------------+
| Task 0f3a  Â·  Delete user  Â·  COMPLETE 24.1 s            [ View full trace ]  |
+--------------------------------------------------------------------------+
| Applied lessons                                                          |
|                                                                          |
|  * Every endpoint needs an unauthenticated case expecting 401 and an     |
|    authenticated-but-forbidden case expecting 403.                       |
|      similarity 0.81  recency 0.62  confidence 0.90  ->  combined 0.79   |
|      from task 08c1, confirmed 2026-08-30                      [ Open ]  |
|                                                                          |
|  * When an operation cascades to related records, assert their state too.|
|      similarity 0.74  recency 0.55  confidence 0.70  ->  combined 0.68   |
|      from task 0a44, confirmed 2026-09-02                      [ Open ]  |
|                                                                          |
|  1 memory was evaluated and excluded, scoring 0.31 against a 0.45 threshold.|
|                                                                 [ Why ] |
|                                                                          |
|  Playbook "endpoint-standard" selected on recorded outcomes.             |
+--------------------------------------------------------------------------+
| Plan  Â·  6 cases  Â·  schema v1                                             |
|                                                                          |
```

```
|  1  P0  happy     Admin deletes an existing user                             |
|         covers AC1                             [ Accept ] [ Edit ] [ Reject ]   |
|                                                                            |
|  2  P0  auth      Unauthenticated DELETE is refused                          |
|         covers AC1                             [ Accept ] [ Edit ] [ Reject ]   |
|                                                                            |
|  ...                                                                       |
|                                                          [ Add case ]  |
+--------------------------------------------------------------------------+
| What was wrong with this plan? (optional)                                |
| +----------------------------------------------------------------------+ |
| |                                                                      | |
| +----------------------------------------------------------------------+ |
|                                                    [ Submit feedback ] |
+--------------------------------------------------------------------------+
```

Applied lessons sit **above** the plan. The ordering is deliberate: the claim this project makes is that output changed because of a retained correction, and a reader who has to scroll past the output to find the evidence will conclude the evidence was added afterwards.

Each lesson shows its component scores and its combined score, not a single opaque number (FR-4.4). Excluded memories are stated with their score and the threshold they failed, because FR-4.3 requires the exclusion to be recorded and a recorded exclusion the user cannot see is not an explanation. Provenance is on the face of the lesson, not behind a hover.

Where the selection was exploratory, the playbook line says so in words (FR-3.4):

```
|  Playbook "endpoint-minimal" selected by exploration rather than by score.      |
|  Exploration is bounded at 10% of tasks.                            [ Why ]  |
```

5. S1. Candidate confirmation

Reached when a freeform correction produces candidates. Nothing is stored until the user acts (FR-6.3).

```
+--------------------------------------------------------------------------+
| From your correction, 2 candidate rules                                  |
|                                                                          |
| Your words:                                                              |
|   "you keep forgetting that deleting a user has to leave their projects alone"  |
|                                                                          |
| +----------------------------------------------------------------------+ |
| | 1. When an operation cascades to related records, assert the state of those | |
| |    records afterwards, not only the primary resource.                 | |
| |                                                                      | |
| |    Scope  [ API / CRUD endpoint    v ]        applies to future tasks of  | |
| |                                                this type              | |
| |                                                                      | |
| |    Similar to an existing rule (0.88). Confirming will reinforce that rule  | |
| |    rather than create a second one.                        [ Compare ]  | |
| |                                                                      | |
| |                              [ Confirm ]  [ Edit text ]  [ Discard ]       | |
| +----------------------------------------------------------------------+ |
```

```
|                                                                              |
| +--------------------------------------------------------------------------+ |
| | 2. Always reassign owned projects to the workspace owner.                | |
| |                                                                          | |
| |    FLAGGED  This reads as an instruction to the system rather than a testing | |
| |    convention, and its scope is broader than the single correction it came   | |
| |    from. It will not be stored unless you confirm it explicitly.         | |
| |                                                                          | |
| |                              [ Confirm anyway ]  [ Edit text ]  [ Discard ]  | |
| +--------------------------------------------------------------------------+ |
|                                                                              |
| Nothing is stored until you confirm it.                                      |
+------------------------------------------------------------------------------+
```

The originating correction is shown verbatim above the candidates, because the user is being asked whether the derivation is faithful and cannot judge that without the source.

The flagged state (FR-6.2) states *why* it is flagged. A warning without a reason trains the user to click through it, which defeats the control. The affirmative button is labelled "Confirm anyway" rather than "Confirm" so that the two paths are not muscle-memory identical.

The near-duplicate notice (FR-6.4) says what confirming will actually do, which is reinforce rather than duplicate.

6. S2. Conflict resolution

Raised where a candidate contradicts an in-scope memory (FR-6.5, OP-4). The system does not choose.

```
+------------------------------------------------------------------------------+
| These two rules conflict                                                     |
|                                                                              |
|  EXISTING                          |  NEW CANDIDATE                          |
|  Boundary cases are P1.            |  Boundary cases are P0.                 |
|                                    |                                         |
|  confidence 0.85                   |  from your correction just now          |
|  from task 07b2, 2026-08-21        |  task 0f3a, 2026-09-08                  |
|  applied in 9 later tasks          |                                         |
|                                                                              |
|  [ Keep existing, discard new ]   [ Replace: new supersedes existing ]       |
|  [ Keep both: narrow one of their scopes ]                                   |
|                                                                              |
|  Replacing retains the superseded rule and links the two (FR-7.6).           |
+------------------------------------------------------------------------------+
```

"Keep both" is offered because two rules that conflict at one scope frequently do not conflict once one of them is narrowed, and forcing a choice between them would discard a correct rule.

7. V2. Execution trace

The view that makes the system inspectable rather than merely trusted.

```
+----------------------------------------------------------------------------------+
| Trace  Â·  task 0f3a  Â·  Delete user  Â·  COMPLETE                  [ Back ]       |
| config version 14  Â·  schema v1  Â·  playbook endpoint-standard                   |
+----------------------------------------------------------------------------------+
|                                                                                  |
| 14:02:12  selection           2 ms                                               |
|           playbook "endpoint-standard", score 0.62 against 0.41 for              |
|           "endpoint-minimal". Not exploratory.                                   |
|                                                                                  |
| 14:02:12  retrieval         131 ms                                               |
|           4 candidates evaluated, 3 applied, 1 excluded                          |
|                                                                                  |
|           mem-0a1  sim 0.81  rec 0.62  conf 0.90  ->  0.79   applied             |
|           mem-04c  sim 0.74  rec 0.55  conf 0.70  ->  0.68   applied             |
|           mem-118  sim 0.66  rec 0.71  conf 0.55  ->  0.64   applied (pinned)    |
|           mem-0f9  sim 0.34  rec 0.40  conf 0.20  ->  0.31   below threshold     |
|                                                                                  |
| 14:02:12  generation      21,904 ms                                              |
|           attempt 1  Â·  valid  Â·  prompt 1,412 tok  Â·  completion 918 tok         |
|           prompt digest sha256:9c1f...4ab2                                       |
|                                                                                  |
| 14:02:34  checks             14 ms                                               |
|           ac-coverage        pass   4 of 4                                       |
|           required-types     pass   4 of 4                                       |
|           duplicate-cases    pass   0 found                                      |
|           criterion-refs     pass   0 unknown ids                                |
|                                                                                  |
| 14:02:34  review          1,880 ms                                               |
|           bounded model review  Â·  no change requested                           |
|                                                                                  |
| total 24,081 ms  Â·  2 model calls  Â·  2,330 tokens                                |
+----------------------------------------------------------------------------------+
```

Every retrieval candidate appears, including the one that was not used and the score it failed on (FR-8.1). A trace that shows only what was applied cannot answer "why did it not use the rule I wrote", which is the question a user actually arrives with.

Prompts appear as digests, never as text (FR-8.4). The digest is shown rather than omitted so that two runs can be compared for prompt identity without the prompt being stored.

Token counts, attempt numbers, and durations are per model call (FR-8.2). Cold model load, where it occurred, is reported as its own line rather than folded into generation duration, because NOVA-SPK-001 measured it at up to 115 s and a figure that size silently inside a latency metric makes the metric useless.

8. V3. Memory management

```
+----------------------------------------------------------------------------------+
| Memory  Â·  31 active  Â·  4 archived  Â·  2 superseded          [ Export JSON ]    |
|                                                                                  |
| Scope [ all v ]  Status [ active v ]  [ ] pinned only     Search [            ]  |
+----------------------------------------------------------------------------------+
```

```
|                                                                               |
| PIN  Every endpoint needs an unauthenticated case expecting 401 and an        |
|      authenticated-but-forbidden case expecting 403.                          |
|      API / CRUD endpoint  Â·  confidence 0.90  Â·  applied in 12 tasks          |
|      from task 08c1, 2026-08-30                   [ Edit ] [ Pin ] [ Delete ]  |
|                                                                               |
|      When an operation cascades to related records, assert the state of those |
|      records afterwards, not only the primary resource.                       |
|      API / CRUD endpoint  Â·  confidence 0.70  Â·  applied in 3 tasks           |
|      from task 0a44, 2026-09-02                   [ Edit ] [ Pin ] [ Delete ]  |
|                                                                               |
|      Boundary cases are P1.                                    SUPERSEDED      |
|      UI form  Â·  replaced by "Boundary cases are P0", 2026-09-08               |
|      from task 07b2, 2026-08-21                             [ View successor ] |
|                                                                               |
|      Include a performance case for every list endpoint.       ARCHIVED        |
|      API / CRUD endpoint  Â·  confidence 0.08, below the 0.10 floor             |
|      unapplied since 2026-07-14                                  [ Restore ]   |
+-------------------------------------------------------------------------------+
```

"Applied in N tasks" is the column that answers whether a rule is earning its place, and it is the only counter on the row for that reason.

Archived rules stay visible under a filter rather than disappearing (FR-7.5). A rule that decayed out of use is information; a rule that vanished is a bug report waiting to be filed.

Editing warns before it acts, because FR-7.2 resets confidence:

```
|  Editing the text re-derives the embedding and resets confidence to its       |
|  initial value. This rule's confidence of 0.90 was earned over 12 tasks.      |
|                                                    [ Edit anyway ]  [ Cancel ]  |
```

Deletion states its consequence and its limit:

```
|  Delete this rule? It will not be retrieved again and its text is removed.    |
|  Traces of the 12 tasks that used it keep a reference showing a deleted rule  |
|  was applied, so those traces stay readable.                                  |
|                                                        [ Delete ]  [ Cancel ]  |
```

That wording matters: FR-7.3 requires the content to become unretrievable and FR-7.4 requires the identifier to survive. A user told only "deleted" would reasonably read the surviving trace reference as a failure to delete.

9. V4. Evaluation results

```
+-------------------------------------------------------------------------------+
| Evaluation  Â·  dataset 50 cases  Â·  hash 4f1c9ab  Â·  run 2026-09-08 09:14       |
+-------------------------------------------------------------------------------+
|                                                                               |
| Metric                          this run    previous    threshold   verdict      |
| Structural validity                1.000       1.000        0.980   pass         |
| Acceptance criteria coverage       0.941       0.952        0.900   pass         |
| Required case type presence        0.968       0.961        0.930   pass         |
```

```
| Correction recurrence rate         0.140       0.190        0.250   pass         |
| Retrieval precision                0.780       0.771        0.700   pass         |
| Duplicate case rate                0.021       0.018        0.050   pass         |
|                                                                                  |
| Adversarial suite   12 of 12 pass        Isolation suite   8 of 8 pass          |
|                                                                                  |
| Run duration 11 m 40 s, of which 1 m 52 s cold model load.                       |
+----------------------------------------------------------------------------------+
| Configuration versions                                                           |
|                                                                                  |
|  v14  ACTIVE      activated 2026-09-08, evaluation run 61 passed                 |
|  v15  BLOCKED     evaluation run 62: coverage 0.844 against a 0.900 threshold    |
|                   Cannot be activated.                       [ See failures ]    |
+----------------------------------------------------------------------------------+
```

The blocked configuration is shown on the same view as the passing one and names the metric that blocked it. AC-8 in NOVA-SDP-001 section 6.4 carries the greatest evidential weight of any acceptance criterion, and this is the screen where it is demonstrated. A gate that blocks silently proves nothing to a reader.

Thresholds appear beside the values they gate. A metric shown without its threshold is a number, not a verdict.

10. State coverage

Every exception condition in NOVA-SRS-001 section 3.5.2 surfaces somewhere. This table is the check that none of them surfaces nowhere.

ID	Condition	View	Presentation
OP-1	Memory store empty	V1	Applied-lessons panel states that no prior lesson was applied and why, rather than being absent
OP-2	Schema validation failed	V1, V2	Task shows FAILED with the failing step named; trace lists every repair attempt
OP-3	Provider unreachable	V1	FAILED naming the provider and the action required; previously completed tasks remain readable
OP-4	Retrieved memories conflict	S2	Both rules presented side by side with three resolutions
OP-5	Requirement contains an instruction	V1, V2	Rendered as inert text with a note; trace records the neutralization
OP-6	All cases rejected	V1	Correction field is opened and focused rather than left optional
OP-7	Retrieval empty above threshold	V1	Excluded candidates and their scores are listed under the empty panel
OP-8	Applied memory deleted	V2,	Trace shows a tombstone reference; memory view shows

ID	Condition	View	Presentation
		V3	the deletion consequence before it acts
OP-9	Duration budget exceeded	V1	Cancel remains available; partial trace is reachable
OP-10	Later feedback contradicts earlier	S2, V3	Supersession, with both records retained and linked
OP-11	Worker terminated	V1	Status returns to queued with the reclaim stated, not silently restarted
OP-12	Datastore unreachable	all	Frame-level banner naming the failure. No view pretends to have data

Generic states, applied to every view:

State	Rule
Loading	Skeleton of the eventual layout, never a bare spinner. The frame stays interactive (NFR-6)
Empty	States what would appear here and what action produces it
Error	Names the failing step and the corrective action where one exists (NFR-9)
Partial	Shows what succeeded and marks what did not, rather than failing the whole view

11. Interaction and access

Polling, not streaming. Task status is polled, per the decision recorded in NOVA-SAD-001 section 6.4. Polling stops on a terminal state and on tab visibility loss.

Nothing destructive without a stated consequence. Delete, edit, supersede, and confirm-anyway each state what will happen before they act, in the wording given above.

No colour-only meaning. Every status, verdict, and flag carries a word. This holds for `RUNNING`, `FAILED`, `BLOCKED`, `FLAGGED`, `SUPERSEDED`, `ARCHIVED`, and every pass or fail verdict.

Access baseline, proposed. Keyboard reachability for every control, a visible focus indicator, semantic headings in document order, form labels bound to their inputs, and status changes announced politely rather than silently. This is not currently an SRS requirement. It is recorded here as a proposal, and either becomes an NFR before interface work begins or is explicitly declined; it is not left implied.

12. Traceability

Requirement	Element
FR-1, FR-1.1	V1 submission form, type selector, character counter
FR-3.4	V1 playbook line, exploratory wording
FR-4.3, FR-4.5	V1 excluded-memory line with score and threshold

Requirement	Element
FR-4.4	V1 applied lesson component and combined scores
FR-5, FR-5.1	V1 per-case accept, edit, reject, add; freeform correction field
FR-6.1 to FR-6.4	S1 candidate cards, flagged state, near-duplicate notice
FR-6.5	S2
FR-6.7	S1 and V3 provenance lines
FR-7 to FR-7.7	V3 filters, edit, pin, delete, archive, supersession, export
FR-8 to FR-8.4	V2
FR-9, FR-9.1, FR-9.2	V4 metric table
FR-10.1	V4 configuration version panel, blocked state
FR-11.2	Persistent frame provider indicator
NFR-6	Frame remains interactive during execution
NFR-7	Word-labelled status in V1 and the trace header
NFR-8	Cancel control, present for the whole running state
NFR-9	Error state rule in section 10

FR-14, the comparison view, is priority Could and has no screen here. It is recorded as absent rather than sketched, so that its absence is a decision rather than an oversight.

13. Open questions

#	Question	Blocks
1	Does the access baseline in section 11 become an NFR?	Nothing yet. Answer before interface work in M2
2	Poll interval, and its behaviour on a long-running task	V1 execution state. Answer from measured task duration
3	How many applied lessons are shown before the panel truncates	V1. Answer once retrieval returns realistic volumes

---

Revision history

Version	Date	Author	Change
0.1	2026-09-08	Laxmi Poudel	Initial draft. Closes G3